Real Time Speed Bump Detection Using Computer Vision

I. INTRODUCTION

India's roads often suffer from **irregularly placed, unmarked, or poorly designed speed breakers**, leading to **sudden vehicle jolts, accidents, and damage**. To tackle this issue, we propose a **machine learning-based system** that can **detect and locate speed breakers** using **image and video data**. By leveraging **computer vision and deep learning**, our model will accurately identify speed bumps and integrate the results into **Google Maps**, providing real-time alerts to drivers. This solution aims to enhance **road safety, navigation, and urban planning**, reducing accident risks caused by unexpected speed breakers.

II. PROBLEM STATEMENT :

- Accidents due to irregular marked & uneven placed speed bumps are significant danger and cause serious injuries and fatalities
- According to a study conducted by the Ministry of Road Transport and Highways nearly 30 accidents occur on a daily basis due to this issue in India
- According to this report accidents due to speed bumps contributes 2.2% share of the total road accidents which is almost 11,084

3. OBJECTIVES AND SCOPE

- Develop a Robust Speed Bump Detection Model: The core objective is to create a machine learning model capable of accurately identifying speed breakers in images or video feeds.

- Accurate Localization of Speed Breakers: Beyond just identifying speed breakers, the system should be able to accurately locate them within the image or video frame.
- Explore Integration Possibilities: Investigate the potential for integrating the system with existing navigation systems or driver-assistance technologies

SCOPE :

The primary goal of this project is to make a ML Model that detects speed bumps and marking it on the google map.

Specifically, we will:

- **Detection of Speed Bumps**: Use pictures and videos to detect presence of speed bumps.
- **Indicating on maps:** Marking Speed Bumps on the maps.
- **Enhanced Road Safety:** Helps drivers anticipate sudden speed breakers, reducing the risk of accidents, vehicle damage, and discomfort.

4. Resources (Hardware & Software)

4.1 Hardware Requirements

Client Side

Processor	Dual Core or above
RAM	4 GB
Disk space	150 GB
Screen Resolution	720p or above
Others	Keyboard, mouse, Internet connection

Developer Side

Processor	Quad Core or above
RAM	8 GB
Disk space	250 GB
Screen Resolution	1080p or above
Others	Keyboard, mouse, Internet connection

4.2 Software Requirements

Client Side

- Web Browser (Google Chrome, Firefox, IE9 or above)
- Windows 7 or above / Linux / Android / IOS

Developer Side

- Operating System: Windows 10 or above /Linux
- Web Browser: Chrome / Firefox / Edge
- IDE: VS Code
- Programming Language: Python
- Runtime Environment: Miniconda
- Backend Framework: Flask
- Database: Sql alchemy
- Version Control System: Git

5. Project Schedule Plan:

The Software Planning objective is to create a flexible framework for estimating resources, costs, and schedules at the project's outset, to be updated regularly. Embracing best and worst-case scenarios, it aims to provide a bounded understanding of project outcomes. More than a visual aid, it is a dynamic narrative, uiding project managers through the timelines, turning complexity into clarity.

Task	16Feb-26Feb	27Feb-8Mar	9Mar-18Mar	19Mar-17Apr	18Apr-27Apr	28Apr-8May
Develop project proposal	11 days					
Analysis		10 days				
Designing			10 days			
Coding				30 days		
Unit Testing					10 days	
Implementation						10 days

Gantt Chart

6. Project Team:

Instructor	:	AREEBA SHAKEEL
Mentor	:	
Project Team Members	:	RAHUL PANDEY, CHITRANSHU SRIVASTAVA

7. Process Description

1. Data Collection Module

Responsible for gathering real and AI-generated images of roads with and without speed bumps from multiple sources like Kaggle, Gemini, and Copilot.

2. Data Preprocessing Module

Handles cleaning, resizing, augmenting, and normalizing images to improve the quality and consistency of the dataset for training.

3. Data Annotation Module

Involves labeling and tagging images manually or using automated tools to classify roads as "with speed bumps" or "without speed bumps."

4. Model Training Module

Focuses on developing and training a deep learning model (e.g., CNN) using the prepared dataset to recognize speed bumps in road images.

5. Model Evaluation Module

Evaluates model performance using metrics such as accuracy, precision, recall, and confusion matrix to ensure effective classification.

6. Model Optimization Module

Applies fine-tuning techniques like hyperparameter tuning, transfer learning, and regularization to improve model accuracy and efficiency.

7. GPS Mapping Module

Integrates GPS data with detected speed bumps and prepares the data for visualization on digital maps for better navigation insights.

8. Web Application Module

Develops a user-friendly web interface using Flask/React to allow users to upload road images and view detection results.

9. Map Visualization Module

Implements interactive maps using Google Maps API or Leaflet.js to display detected speed bumps with markers and other relevant data.

10. Image Upload and Processing Module

Enables users to upload road images via the web or mobile application and process them in real time to detect speed bumps.

11. Security and Authentication Module

Ensures secure access to the web and mobile application by implementing authentication methods such as JWT (JSON Web Tokens) and role-based access control (RBAC).

12. Database Management Module

Stores processed images, detection results, GPS coordinates, and user data securely using databases like MySQL, PostgreSQL, or Firebase for efficient retrieval and analysis.

8. Contribution of the student in the project:

i. **Rahul : Data Collection Module, Data Annotation Module, Model Evaluation Module, GPS Mapping Module, Map Visualization Module, Security and Authentication Module**

ii. **Chitranshu : Data Preprocessing Module, Model Training Module, Model Optimization Module, Web Application Module, Image Upload and Processing Module, Database Management Module**

9. Conclusion

BumpSafe Vision is a solution for speed bumps detection and locating it on the map using ML algorithms and map related APIs. It tends to improve road safety by avoiding sudden braking and collision..